

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006, as amended by Commission Regulation (EU) 2020/878

Version 7.10
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GENERIC EU MSDS - NO COUNTRY SPECIFIC DATA - NO OEL DATA

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifiers

Product name : Hydrogen peroxide 30% (stabilized) for synthesis

Product Number : 8.22287

Catalogue No. : 822287

Brand : Millipore

UFI : YGRX-T5D9-Q994-UDS6

REACH No. : 01-2119485845-22-XXXX

1.2 Relevant identified uses of the substance or mixture and uses advised against

Identified uses : Chemical for synthesis

Uses advised against : Reserved for industrial and professional use.

1.3 Details of the supplier of the safety data sheet

Company : Sigma-Aldrich Israel Ltd. Tel Aviv
3 PARK RABIN, PLAUT
76100 REHOVOT
ISRAEL

Telephone : +972 972 8948-4222

Fax : +972 972 8948-4200

1.4 Emergency telephone number

Emergency Phone # : +972 3-763-0639 (CHEMTREC)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Serious eye damage, Category 1 H318: Causes serious eye damage.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Millipore- 8.22287

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Hazard pictograms

:



Signal word

: Danger

Hazard statements

: H318 Causes serious eye damage.

Precautionary statements

: **Prevention:**

P280 Wear eye protection/ face protection.

Response:

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Hazardous components which must be listed on the label:

Hydrogen Peroxide

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
Hydrogen Peroxide	7722-84-1 231-765-0 008-003-00-9 01-2119485845- 22-xxxx	Ox. Liq. 1; H271 Acute Tox. 4; H302 Acute Tox. 4; H332	>= 30 - < 35



		Skin Corr. 1A; H314 Eye Dam. 1; H318 STOT SE 3; H335 (Respiratory system) Aquatic Chronic 3; H412	
		specific concentration limit Ox. Liq. 1; H271 >= 70 % Ox. Liq. 2; H272 50 - < 70 % Skin Corr. 1A; H314 >= 70 % Skin Corr. 1B; H314 50 - < 70 % Skin Irrit. 2; H315 35 - < 50 % Eye Dam. 1; H318 8 - < 50 % Eye Irrit. 2; H319 5 - < 8 % STOT SE 3; H335 >= 35 %	
		Acute toxicity estimate Acute oral toxicity: 693,7 mg/kg Acute inhalation toxicity (vapour): 11,1 mg/l	

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

If inhaled : After inhalation: fresh air.

In case of skin contact : In case of skin contact: Take off immediately all contaminated clothing. Rinse skin with water/ shower.



In case of eye contact : After eye contact: rinse out with plenty of water.
Immediately call in ophthalmologist.
Remove contact lenses.

If swallowed : After swallowing: immediately make victim drink
water (two glasses at most).
Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in the labelling (see section 2.2) and/or in section 11

4.3 Indication of any immediate medical attention and special treatment needed

No data available

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Use extinguishing measures that are appropriate to
local circumstances and the surrounding environment.

Unsuitable extinguishing media : For this substance/mixture no limitations of
extinguishing agents are given.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire fighting : Not combustible.

Hazardous combustion products : Nature of decomposition products not known.

5.3 Advice for firefighters

Special protective equipment for fire-fighters : Stay in danger area only with self-contained breathing
apparatus. Prevent skin contact by keeping a safe
distance or by wearing suitable protective clothing.

Further information : Prevent fire extinguishing water from contaminating
surface water or the ground water system.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Advice for non-emergency personnel:
Do not breathe vapours, aerosols.
Avoid substance contact.
Ensure adequate ventilation.
Evacuate the danger area, observe emergency
procedures, consult an expert.

For personal protection see section 8.

6.2 Environmental precautions

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Canada



Environmental precautions : Do not empty into drains.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Cover drains. Collect, bind, and pump off spills.

Observe possible material restrictions (see sections 7 and 10).

Take up with liquid-absorbent and neutralising material (e.g. Chemizorb® H⁺, Merck Art. No. 101595). Dispose of properly. Clean up affected area.

6.4 Reference to other sections

For disposal considerations see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Observe label precautions.

Hygiene measures : Immediately change contaminated clothing. Apply preventive skin protection. Wash hands and face after working with substance.

For precautions see section 2.2.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : No metal containers.

Close containers in such a way to enable internal pressure to escape (e.g. excess pressure valve).

Further information on storage conditions : Tightly closed. Protected from light. Do not store near combustible materials.

Storage class (TRGS 510) : 5.1B, Oxidizing hazardous materials

Further information on storage stability : Recommended storage temperature see product label.

Recommended storage temperature : Recommended storage temperature see product label.

7.3 Specific end use(s)

Apart from the uses mentioned in section 1.2 no other specific uses are stipulated



SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Derived No Effect Level (DNEL) according to Regulation (EC) No. 1907/2006

Substance name	End Use	Exposure routes	Potential health effects	Value
Hydrogen Peroxide	Worker DNEL, acute	Inhalation	Local effects	3 mg/m ³
	Worker DNEL, longterm	Inhalation	Local effects	1,4 mg/m ³
	Consumer DNEL, acute	Inhalation	Local effects	1,93 mg/m ³
	Consumer DNEL, longterm	Inhalation	Local effects	0,21 mg/m ³

Predicted No Effect Concentration (PNEC) according to Regulation (EC) No. 1907/2006

Substance name	Environmental Compartment	Value
Hydrogen Peroxide	Fresh water	0,0126 mg/l
	Marine water	0,0126 mg/l
	Aquatic intermittent release	0,0138 mg/l
	Sewage treatment plant	4,66 mg/l
	Fresh water sediment	0,47 mg/kg
	Marine sediment	0,47 mg/kg
	Soil	0,0023 mg/kg

8.2 Exposure controls

Individual protection measures, such as personal protective equipment (PPE)

Eye/face protection : Tightly fitting safety goggles

Hand protection

Material : Latex gloves
Break through time : 480 min
Glove thickness : 0,6 mm
Protective index : Full contact
Manufacturer : Lapren® (KCL 706 / Aldrich Z677558, Size M)

Material : Nitrile rubber
Break through time : 480 min
Glove thickness : 0,11 mm
Protective index : Splash contact
Manufacturer : KCL 741 Dermatril® L

Remarks : This recommendation applies only to the product stated in the safety data sheet, supplied by us and for the designated use. When dissolving in or mixing with other substances and under conditions deviating from those stated in EN 16523-1 please contact the supplier of CE-approved gloves (e.g. KCL GmbH, D-



Skin and body protection : protective clothing

Respiratory protection : required when vapours/aerosols are generated.
Our recommendations on filtering respiratory protection are based on the following standards: DIN EN 143, DIN 14387 and other accompanying standards relating to the used respiratory protection system.

Recommended Filter : filter NO
type:

The entrepreneur has to ensure that maintenance, cleaning and testing of respiratory protective devices are carried out according to the instructions of the producer.

These measures have to be properly documented.

Protective measures : protective clothing

Environmental exposure controls

Advice : Do not empty into drains.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state : liquid

Color : colourless

Odor : slight

Melting point/ range : -26 °C

Boiling point/boiling range : 107 °C (1,013 hPa)

Flammability : No data available

Upper explosion limit /
Upper flammability limit : No data available

Lower explosion limit /
Lower flammability limit : No data available

Flash point : not combustible

Autoignition temperature : No data available



Decomposition temperature	: > 100 °C
pH	: 2 - 4 (20 °C) Concentration: 100 %
Viscosity, dynamic	: No data available
Viscosity, kinematic	: No data available
Flow time	: No data available
Water solubility	: No data available
Partition coefficient: n-octanol/water	: No data available
Vapor pressure	: ca. 18 hPa (20 °C)
Relative density	: No data available
Density	: 1,11 g/cm ³ (20 °C)
Relative vapour density	: No data available

9.2 Other information

Explosives	: Not classified as explosive.
Oxidizing properties	: Oxidizing potential
Burning rate	: No data available
Evaporation rate	: No data available

SECTION 10: Stability and reactivity

10.1 Reactivity

Has a fire-promoting effect due to release of oxygen.

10.2 Chemical stability

heat-sensitive

Sensitivity to light

Contains the following stabiliser(s):
Disodium pyrophosphate(0,015 %)



phosphoric acid(0,01 %)
Ammonium nitrate(0,006 %)

10.3 Possibility of hazardous reactions

Hazardous reactions : Risk of explosion with:

Acetaldehyde
Acetone
Activated charcoal
Alcohols
formic acid
Ammonia
combustible substances
vinyl acetate
Organic Substances
Powdered metals
Dust
hydrazine and derivatives
hydrides
Ether
Potassium
anilines
Metallic salts
acetic acid
Acetic anhydride
Formaldehyde
furfuryl alcohol
oils
sodium
Lithium
lithium aluminium hydride
organic solvents
Magnesium
metallic oxides
Methanol
Reducing agents
Oxides of phosphorus
butanol
with
Sulphuric acid
alkali hydroxides
with
Heavy metals
Exothermic reaction with:

alkali hydroxides
antimony sulfide
tin (II) chloride
Sulphides
3-BROMO-5-CHLORO-4-HYDROXYBENZALDEHYDE
nitric acid (conc.)
ethanol
glycerol
Potassium hydroxide



phosphorus
metallic oxides
Sodium hydroxide
Aldehydes
nonmetals
nonmetallic oxides
strong alkalis
Amines
Acids
Oxidizing agents
alkali salts
Alkali metals
Alkaline earth metals
iodides
peroxi compounds
Brass
organic nitro compounds
phenol
with
metal catalysts
Risk of ignition or formation of inflammable gases or
vapours with:
potassium permanganate
Wood/Sawdust
vinyl acetate
with
Catalyst

10.4 Conditions to avoid

Conditions to avoid : Heating.

10.5 Incompatible materials

Materials to avoid : Metals

10.6 Hazardous decomposition products

In the event of fire: see section 5

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Mixture

Acute toxicity

Acute toxicity estimate Oral - > 2.000 mg/kg

(Calculation method)

Acute toxicity estimate Inhalation - 4 h - > 20 mg/l - vapour(Calculation method)

Dermal: No data available

Skin corrosion/irritation

Remarks: After long-term exposure to the chemical:

Causes skin burns.



Serious eye damage/eye irritation

Remarks: conjunctivitis

Respiratory or skin sensitization

No data available

Germ cell mutagenicity

No data available

Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

No data available

Specific target organ toxicity - repeated exposure

No data available

Aspiration hazard

No data available

11.2 Additional Information**Endocrine disrupting properties****Product:**

Assessment

The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Dizziness

Unconsciousness

Diarrhoea

Nausea

Vomiting

Headache

Convulsions

muscle twitching

insomnia

shock

Irritation and corrosion

conjunctivitis

Risk of serious damage to eyes.

Other dangerous properties can not be excluded.

Handle in accordance with good industrial hygiene and safety practice.



Components

Hydrogen Peroxide

Acute toxicity

LD50 Oral - Rat - female - 693,7 mg/kg

(OECD Test Guideline 401)

Acute toxicity estimate Oral - 693,7 mg/kg

(ATE value derived from LD50/LC50 value)

Acute toxicity estimate Inhalation - 4 h - 11,1 mg/l - vapour

(Expert judgement)

LD50 Dermal - Rabbit - male and female - > 2.000 mg/kg

(US-EPA)

Skin corrosion/irritation

Remarks: Causes severe burns.

Classified according to Regulation (EU) 1272/2008, Annex VI (Table 3.1/3.2)

Serious eye damage/eye irritation

Remarks: Causes serious eye damage.

Respiratory or skin sensitization

No data available

Germ cell mutagenicity

Method: OECD Test Guideline 474

Species: Mouse - male and female - Bone marrow

Result: negative

Carcinogenicity

No data available

Reproductive toxicity

No data available

Specific target organ toxicity - single exposure

Inhalation - May cause respiratory irritation. - Respiratory Tract

Specific target organ toxicity - repeated exposure

Aspiration hazard

No data available

SECTION 12: Ecological information

12.1 Toxicity

Product:

Toxicity to fish : Remarks: No data available

Ecotoxicology Assessment

Chronic aquatic toxicity : This product has no known ecotoxicological effects.

Components:

Hydrogen Peroxide:

Toxicity to fish : LC50 (Pimephales promelas (fathead minnow)): 16,4



	mg/l End point: mortality Exposure time: 96 h Test Type: semi-static test Analytical monitoring: yes Method: US-EPA
Toxicity to daphnia and other aquatic invertebrates	: LC50 (<i>Daphnia pulex</i> (Water flea)): 2,4 mg/l End point: mortality Exposure time: 48 h Test Type: semi-static test Analytical monitoring: yes Method: US-EPA
Toxicity to algae/aquatic plants	: ErC50 (<i>Skeletonema costatum</i> (marine diatom)): 1,38 mg/l Exposure time: 72 h Test Type: static test Analytical monitoring: yes GLP: yes Remarks: (ECHA) NOEC (<i>Skeletonema costatum</i> (marine diatom)): 0,63 mg/l Exposure time: 72 h Test Type: static test Analytical monitoring: yes GLP: yes Remarks: (ECHA)
Toxicity to microorganisms	: EC50 (activated sludge): 466 mg/l Exposure time: 30 min Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 209 GLP: yes EC50 (activated sludge): > 1.000 mg/l Exposure time: 3 h Test Type: static test Analytical monitoring: yes Method: OECD Test Guideline 209 GLP: yes
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	: NOEC: 0,63 mg/l End point: reproduction rate Exposure time: 21 d Species: <i>Daphnia magna</i> (Water flea) Test Type: flow-through test GLP: yes Remarks: (ECHA)



12.2 Persistence and degradability

Product:

Biodegradability : Result: Readily biodegradable.

Remarks: No data available

Components:

Hydrogen Peroxide:

Biodegradability : Test Type: aerobic
Inoculum: activated sludge
Result: Readily biodegradable.
Biodegradation: > 99 %
Exposure time: 0,5 h
GLP: yes
Remarks: (ECHA)

12.3 Bioaccumulative potential

Product:

Bioaccumulation : Remarks: No data available

Components:

Hydrogen Peroxide:

Bioaccumulation : Remarks: No data available

12.4 Mobility in soil

Product:

Stability in soil : Remarks: No data available

12.5 Results of PBT and vPvB assessment

Product:

Assessment : This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Components:

Hydrogen Peroxide:

Assessment : Substance does not meet the criteria for PBT or vPvB according to Regulation (EC) No 1907/2006, Annex XIII.

12.6 Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components



considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

12.7 Other adverse effects

Product:

Additional ecological information : No interference with wastewater treatment plants are to be expected when used properly.

Discharge into the environment must be avoided.

No data available

Components:

Hydrogen Peroxide:

Additional ecological information : No interference with wastewater treatment plants are to be expected when used properly.

Discharge into the environment must be avoided.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Product : Waste material must be disposed of in accordance with the national and local regulations. Leave chemicals in original containers. No mixing with other waste. Handle uncleaned containers like the product itself.

SECTION 14: Transport information

14.1 UN number or ID number

ADR : UN 2014

IMDG : UN 2014

IATA : UN 2014

14.2 UN proper shipping name

ADR : HYDROGEN PEROXIDE, AQUEOUS SOLUTION

IMDG : HYDROGEN PEROXIDE, AQUEOUS SOLUTION

IATA : Hydrogen peroxide, aqueous solution

14.3 Transport hazard class(es)



	Class	Subsidiary risks
ADR	: 5.1	8
IMDG	: 5.1	8
IATA	: 5.1	8

14.4 Packing group

ADR

Packing group	: II
Classification Code	: OC1
Hazard Identification Number	: 58
Labels	: 5.1 (8)
Tunnel restriction code	: (E)

IMDG

Packing group	: II
Labels	: 5.1 (8)
EmS Code	: F-H, S-Q

IATA (Cargo)

Packing instruction (cargo aircraft)	: 554
Packing instruction (LQ)	: Y540
Packing group	: II
Labels	: Division 5.1 - Oxidizing substances, Class 8 - Corrosive substances

IATA_P (Passenger)

Packing instruction (passenger aircraft)	: 550
Packing instruction (LQ)	: Y540
Packing group	: II
Labels	: Division 5.1 - Oxidizing substances, Class 8 - Corrosive substances

14.5 Environmental hazards

ADR

Environmentally hazardous	: no
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IMDG

Marine pollutant	: no
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14.6 Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable for product as supplied.



SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

REACH - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles (Annex XVII) : Conditions of restriction for the following entries should be considered:
Number on list 3

REACH - Candidate List of Substances of Very High Concern for Authorisation (Article 59). : Not applicable

Regulation (EU) No 2024/590 on substances that deplete the ozone layer : Not applicable

Regulation (EU) 2019/1021 on persistent organic pollutants (recast) : Not applicable

REACH - List of substances subject to authorisation (Annex XIV) : Not applicable

Regulation (EU) 2019/1148 on the marketing and use of explosives precursors

Acquisition, introduction, possession or use of this product by the general public is restricted by Regulation (EU) 2019/1148. All suspicious transactions, and significant disappearances and thefts should be reported to the relevant national contact point. Hydrogen Peroxide (ANNEX I)

This product is regulated by Regulation (EU) 2019/1148: all suspicious transactions, and significant disappearances and thefts should be reported to the relevant national contact point. Ammonium nitrate (ANNEX I) (ANNEX I)

P8

15.2 Chemical safety assessment

A Chemical Safety Assessment has been carried out for this substance.

SECTION 16: Other information

Full text of H-Statements

H271 : May cause fire or explosion; strong oxidizer.
H302 : Harmful if swallowed.
H314 : Causes severe skin burns and eye damage.
H318 : Causes serious eye damage.
H332 : Harmful if inhaled.

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H335	: May cause respiratory irritation.
H412	: Harmful to aquatic life with long lasting effects.
H271	: May cause fire or explosion; strong oxidizer.
H272	: May intensify fire; oxidizer.
H314	: Causes severe skin burns and eye damage.
H315	: Causes skin irritation.
H318	: Causes serious eye damage.
H319	: Causes serious eye irritation.

Full text of other abbreviations

Acute Tox.	: Acute toxicity
Aquatic Chronic	: Long-term (chronic) aquatic hazard
Eye Dam.	: Serious eye damage
Ox. Liq.	: Oxidizing liquids
Skin Corr.	: Skin corrosion
STOT SE	: Specific target organ toxicity - single exposure
Ox. Liq.	: Oxidizing liquids

Skin Corr.	: Skin corrosion
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Skin Irrit.	: Skin irritation
Eye Dam.	: Serious eye damage
Eye Irrit.	: Eye irritation
STOT SE	: Specific target organ toxicity - single exposure

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations



concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information : The information is believed to be correct but is not exhaustive and will be used solely as a guideline, which is based on current knowledge of the chemical substance or mixture and is applicable to appropriate safety precautions for the product. It does not represent any guarantee of the properties of the product. Sigma-Aldrich Corporation and its Affiliates shall not be held liable for any damage resulting from handling or from contact with the above product. See www.sigma-aldrich.com and/or the reverse side of invoice or packing slip for additional terms and conditions of sale.
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Classification of the mixture:

Eye Dam. 1 H318

Classification procedure:

Calculation method

REG_EU / EN



Annex: Exposure scenario

Identified uses:

Use: Industrial use

SU 3: Industrial uses: Uses of substances as such or in preparations at industrial sites
SU 3, SU 10: Industrial uses: Uses of substances as such or in preparations at industrial sites, Formulation [mixing] of preparations and/ or re-packaging (excluding alloys)
PC19: Intermediate
PC39: Cosmetics, personal care products
PROC1: Use in closed process, no likelihood of exposure
PROC2: Use in closed, continuous process with occasional controlled exposure
PROC3: Use in closed batch process (synthesis or formulation)
PROC4: Use in batch and other process (synthesis) where opportunity for exposure arises
PROC5: Mixing or blending in batch processes for formulation of preparations and articles (multistage and/ or significant contact)
PROC8a: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at non-dedicated facilities
PROC8b: Transfer of substance or preparation (charging/ discharging) from/ to vessels/ large containers at dedicated facilities
PROC9: Transfer of substance or preparation into small containers (dedicated filling line, including weighing)
PROC10: Roller application or brushing
PROC14: Production of preparations or articles by tableting, compression, extrusion, pelletisation
PROC15: Use as laboratory reagent
ERC1, ERC2, ERC4, ERC6a, ERC6b: Manufacture of substances, Formulation of preparations, Industrial use of processing aids in processes and products, not becoming part of articles, Industrial use resulting in manufacture of another substance (use of intermediates), Industrial use of reactive processing aids

Use: Professional use

SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
SU 22: Professional uses: Public domain (administration, education, entertainment, services, craftsmen)
PC39: Cosmetics, personal care products
ERC8a, ERC8d: Wide dispersive indoor use of processing aids in open systems, Wide dispersive outdoor use of processing aids in open systems

Use: Consumer use

SU 21: Consumer uses: Private households (= general public = consumers)
SU 21: Consumer uses: Private households (= general public = consumers)
PC39: Cosmetics, personal care products
ERC8a, ERC8d: Wide dispersive indoor use of processing aids in open systems, Wide dispersive outdoor use of processing aids in open systems



1. Short title of Exposure Scenario: Industrial use

Main User Groups	: SU 3
Sectors of end-use	: SU 3, SU 10
Chemical product category	: PC19, PC39
Process categories	: PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC10, PROC14, PROC15
Environmental Release Categories	: ERC1, ERC2, ERC4, ERC6a, ERC6b:

2. Exposure scenario

2.1 Contributing scenario controlling environmental exposure for: ERC1, ERC2, ERC4, ERC6a, ERC6b

Amount used

Annual amount per site	: 1010 t
Remarks	: (refers to pure substance)

Other given operational conditions affecting environmental exposure

Number of emission days per year	: 360
Emission or Release Factor: Air	: 0,10 %
Emission or Release Factor: Water	: 0,50 %
Emission or Release Factor: Soil	: 0,10 %

Technical conditions and measures / Organizational measures

Air	: Use of air emission abatement equipments.
Water	: Biological waste water treatment plant

Conditions and measures related to municipal sewage treatment plant

Type of Sewage Treatment Plant	: Municipal sewage treatment plant
Flow rate of sewage treatment plant effluent	: 2.000 m3/d
Percentage removed from waste water	: 97 %

2.2 Contributing scenario controlling worker exposure for: PROC1, PROC2, PROC3, PROC8b, PROC15

Product characteristics

Concentration of the Substance in Mixture/Article	: Covers the percentage of the substance in the product up to 70 %.
Physical Form (at time of use)	: Medium volatile liquid
Process Temperature	: < 70 °C

Frequency and duration of use

Frequency of use	: 8 hours/day
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Other operational conditions affecting workers exposure

Outdoor / Indoor	: Indoor with local exhaust ventilation (LEV)
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Organisational measures to prevent /limit releases, dispersion and exposure
Covers daily exposures up to 8 hours.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear suitable gloves (tested to EN374) and eye protection.

2.3 Contributing scenario controlling worker exposure for: PROC4

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 70 %.
Physical Form (at time of use) : Medium volatile liquid
Process Temperature : < 70 °C

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor with LEV and good general ventilation

Organisational measures to prevent /limit releases, dispersion and exposure
Covers daily exposures up to 8 hours.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear suitable gloves (tested to EN374) and eye protection.

2.4 Contributing scenario controlling worker exposure for: PROC5, PROC8a, PROC9

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 70 %.
Physical Form (at time of use) : Medium volatile liquid
Process Temperature : < 70 °C

Frequency and duration of use

Frequency of use : < 4 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor with LEV and enhanced general ventilation

Organisational measures to prevent /limit releases, dispersion and exposure
Avoid carrying out operation for more than 4 hours.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear suitable gloves (tested to EN374) and eye protection.

2.5 Contributing scenario controlling worker exposure for: PROC10, PROC14

Product characteristics

Concentration of the Substance in Mixture/Article : Covers the percentage of the substance in the product up to 70 %.
Physical Form (at time of use) : Medium volatile liquid
Process Temperature : < 70 °C

Frequency and duration of use

Frequency of use : 8 hours/day

Other operational conditions affecting workers exposure

Outdoor / Indoor : Indoor with LEV and enhanced general ventilation



Organisational measures to prevent /limit releases, dispersion and exposure
Covers daily exposures up to 8 hours.

Conditions and measures related to personal protection, hygiene and health evaluation

Wear suitable gloves (tested to EN374) and eye protection.

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR*
ERC1	EUSES		Fresh water			0,61
ERC2	EUSES		Fresh water			0,61
ERC4	EUSES		Fresh water			0,61
ERC6a	EUSES		Fresh water			0,61
ERC6b	EUSES		Fresh water			0,61

Workers

Contributing Scenario	Exposure Assessment Method	Specific conditions	Value	Level of Exposure	RCR*
PROC1	ECETOC TRA, modified	longterm, inhalative, systemic			< 0,01
PROC2	ECETOC TRA, modified	longterm, inhalative, systemic			0,35
PROC3	ECETOC TRA, modified	longterm, inhalative, systemic			0,71
PROC8b	ECETOC TRA, modified	longterm, inhalative, systemic			0,89
PROC15	ECETOC TRA, modified	longterm, inhalative, systemic			0,71

*Risk characterisation ratio

PROC4	ECETOC TRA, modified	longterm, inhalative, systemic			0,99
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*Risk characterisation ratio

PROC5	ECETOC TRA, modified	longterm, inhalative, systemic			0,64
PROC8a	ECETOC TRA, modified	longterm, inhalative, systemic			0,64
PROC9	ECETOC TRA,	longterm,			0,64



	modified	inhalative, systemic			
*Risk characterisation ratio					
PROC10	ECETOC TRA, modified	longterm, inhalative, systemic			0,91
PROC14	ECETOC TRA, modified	longterm, inhalative, systemic			0,91

*Risk characterisation ratio

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

For scaling of worker exposure assessments performed with ECETOC TRA, please consult the Merck tool ScIDeEx® at www.merckmillipore.com/scideex.
Please refer to the following documents: ECHA Guidance on information requirements and chemical safety assessment Chapter R.12: Use descriptor system; ECHA Guidance for downstream users; ECHA Guidance on information requirements and chemical safety assessment Part D: Exposure Scenario Building, Part E: Risk Characterisation and Part G: Extending the SDS; VCI/Cefic REACH Practical Guides on Exposure Assessment and Communications in the Supply Chain; CEFIC Guidance Specific Environmental Release Categories (SPERCs).

1. Short title of Exposure Scenario: Professional use

Main User Groups : **SU 22**
Sectors of end-use : **SU 22**
Chemical product category : **PC39**
Environmental Release Categories : **ERC8a, ERC8d:**

2. Exposure scenario

2.1 Contributing scenario controlling environmental exposure for: ERC8a, ERC8d

Amount used

Annual amount per site : 185 t
Remarks : (refers to pure substance)

Other given operational conditions affecting environmental exposure

Number of emission days per year : 360
Emission or Release Factor: Air : 10 %
Emission or Release Factor: Water : 5 %



Emission or Release Factor: Soil : 8 %

3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR*
ERC8a	EUSES		Fresh water			0,94
ERC8d	EUSES		Fresh water			0,94

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

Please refer to the following documents: ECHA Guidance on information requirements and chemical safety assessment Chapter R.12: Use descriptor system; ECHA Guidance for downstream users; ECHA Guidance on information requirements and chemical safety assessment Part D: Exposure Scenario Building, Part E: Risk Characterisation and Part G: Extending the SDS; VCI/Cefic REACH Practical Guides on Exposure Assessment and Communications in the Supply Chain; CEFIC Guidance Specific Environmental Release Categories (SPERCs).

1. Short title of Exposure Scenario: Consumer use

Main User Groups : **SU 21**
Sectors of end-use : **SU 21**
Chemical product category : **PC39**
Environmental Release Categories : **ERC8a, ERC8d:**

2. Exposure scenario

2.1 Contributing scenario controlling environmental exposure for: ERC8a, ERC8d

Amount used

Annual amount per site : 185 t
Remarks : (refers to pure substance)

Other given operational conditions affecting environmental exposure

Number of emission days per year : 360
Emission or Release Factor: Air : 10 %
Emission or Release Factor: Water : 5 %
Emission or Release Factor: Soil : 8 %



3. Exposure estimation and reference to its source

Environment

Contributing Scenario	Exposure Assessment Method	Specific conditions	Compartment	Value	Level of Exposure	RCR*
ERC8a	EUSES		Fresh water			0,94
ERC8d	EUSES		Fresh water			0,94

4. Guidance to Downstream User to evaluate whether he works inside the boundaries set by the Exposure Scenario

Please refer to the following documents: ECHA Guidance on information requirements and chemical safety assessment Chapter R.12: Use descriptor system; ECHA Guidance for downstream users; ECHA Guidance on information requirements and chemical safety assessment Part D: Exposure Scenario Building, Part E: Risk Characterisation and Part G: Extending the SDS; VCI/Cefic REACH Practical Guides on Exposure Assessment and Communications in the Supply Chain; CEFIC Guidance Specific Environmental Release Categories (SPERCs).

